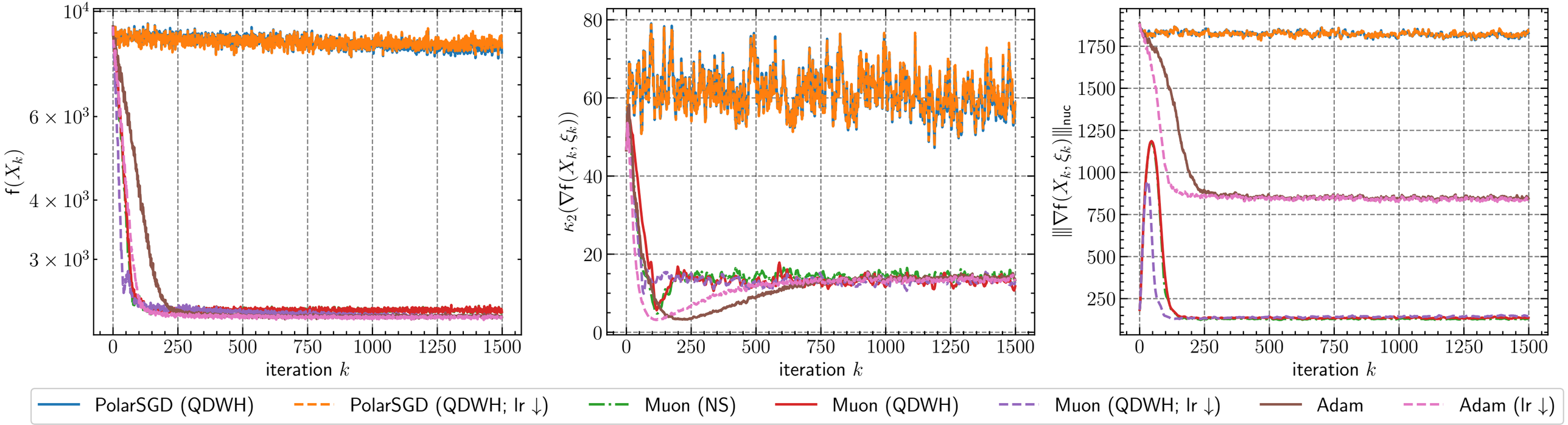
$10^4$
$6 \times 10^3$
$4 \times 10^3$
$3 \times 10^3$
$\mathsf{f}(X_k)$
0
250
500
750
1000
1250
1500
iteration $k$
80
60
40
20
0
$\kappa_2(\nabla \mathsf{f}(X_k, \xi_k))$
iteration $k$
1750
1500
1250
1000
750
500
250
$|||\nabla \mathsf{f}(X_k, \xi_k)|||_{\mathsf{nuc}}$
iteration $k$
PolarSGD (QDWH)
PolarSGD (QDWH; lr ↓)
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)